Use mental multiplication and division strategies

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Calculate 25 x 3,600 mentally. Copy or complete the first step.

2. Calculate 4,800 / 16 mentally.

3. Calculate 125 x 64 using factors.

4. Miro says: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate 25 x 3,600 mentally. Copy or complete the first step.
   Answer 90,000.

2. Calculate 4,800 / 16 mentally.
   Answer 300.

3. Calculate 125 x 64 using factors.
   Answer 8,000.

4. Miro says: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100. Is this correct? Explain.
   Answer Since 25 = 100 / 4, divide by 4 and then multiply by 100.

Use mental multiplication and division strategies

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Calculate 25 x 3,600 mentally.

2. Calculate 4,800 / 16 mentally.

3. Calculate 125 x 64 using factors.

4. Solve this independently and show every stage: Calculate 25 x 3,600 mentally.

5. Use a second method or representation for: Calculate 4,800 / 16 mentally.

6. Check this result using an inverse, estimate or known fact: Calculate 125 x 64 using factors.

7. Find three efficient mental methods for 48 x 250.

8. Identify and correct this misconception: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Calculate 125 x 64 using factors.

Teacher answers and checking prompts

1. Calculate 25 x 3,600 mentally.
   Answer 90,000.

2. Calculate 4,800 / 16 mentally.
   Answer 300.

3. Calculate 125 x 64 using factors.
   Answer 8,000.

4. Solve this independently and show every stage: Calculate 25 x 3,600 mentally.
   Answer 90,000.

5. Use a second method or representation for: Calculate 4,800 / 16 mentally.
   Answer Expected result: 300. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Calculate 125 x 64 using factors.
   Answer Expected result: 8,000. A valid check must be shown.

7. Find three efficient mental methods for 48 x 250.
   Answer 12,000. Methods may use 250 = 1,000 / 4 or 48 / 4 x 1,000.

8. Identify and correct this misconception: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100.
   Answer Since 25 = 100 / 4, divide by 4 and then multiply by 100.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Calculate 125 x 64 using factors.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Use mental multiplication and division strategies

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ______________

1. Find three efficient mental methods for 48 x 250.

2. Prove the answer to this question, rather than only calculating it: Calculate 25 x 3,600 mentally.

3. Solve this in two different ways and compare them: Calculate 4,800 / 16 mentally.

4. Work backwards from the answer to reconstruct the question: 8,000.

5. Explain why this reasoning fails: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100.

6. Create a new example that tests the same idea as: Calculate 125 x 64 using factors.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Find three efficient mental methods for 48 x 250.
   Answer 12,000. Methods may use 250 = 1,000 / 4 or 48 / 4 x 1,000.

2. Prove the answer to this question, rather than only calculating it: Calculate 25 x 3,600 mentally.
   Answer Expected result: 90,000. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Calculate 4,800 / 16 mentally.
   Answer Expected result: 300. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 8,000.
   Answer One valid reconstruction is: Calculate 125 x 64 using factors. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro divides by 4 when multiplying by 25 but forgets to multiply by 100.
   Answer Since 25 = 100 / 4, divide by 4 and then multiply by 100.

6. Create a new example that tests the same idea as: Calculate 125 x 64 using factors.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.